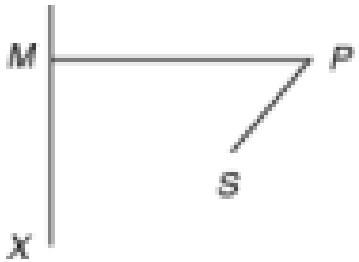
Conic Sections
_1. Parabola _1

Definition

If a point moves in a plane such that its distance from a fixed point bears a constant ratio to its perpendicular distance from a fixed straight line then the path described by the moving point is called a conic. In other words, if S is a fixed point, l is a fixed straight line and P is a moving point and PM is the perpendicular distance from P on l, such that

$$\frac{SP}{PM} = \text{constant},$$

then the locus of P is called a conic. This constant is called the eccentricity of the conic and is denoted by e.



If $e = 1$, the conic is called a parabola.

If $e < 1$, the conic is called an ellipse.

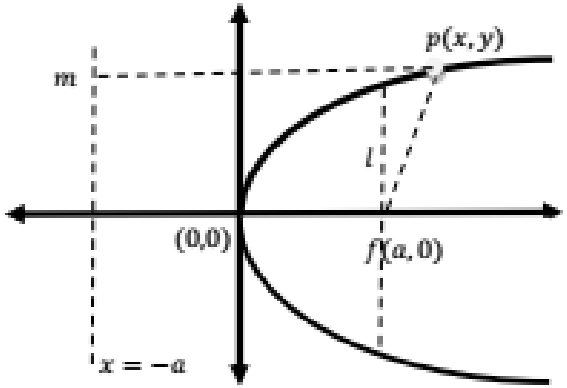
If $e > 1$, the conic is called a hyperbola.

The fixed-point S is called the focus of the conic. The fixed straight line is called the directrix of the conic.

Parabola

Equation of parabola:

The equation of a parabola whose vertex is the origin and whose axis is the positive side of the x-axis



$\because \; pf = pm$

$\therefore \sqrt{(x-a)^2 + (y-0)^2} = x + a$

$(x-a)^2 + (y-0)^2 = (x+a)^2$

$x^2 - 2a\,x + a^2 + y^2 = x^2 + 2a\,x + a^2$

$\therefore y^2 = 4a\,x$

Length of latus rectum:

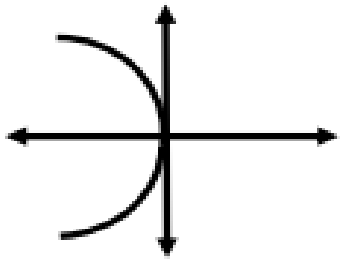
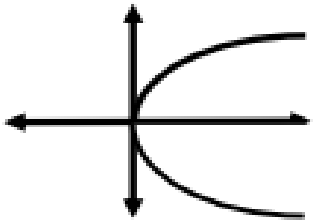
Assume that the length of latus rectum is $2l$

$\because$ The point (a, l) lies on the parabola

$\therefore \; l^2 = 4a \times a = 4\,a^2 \quad \longrightarrow \quad l = 2a \quad \Longrightarrow \quad 2l = 4a$

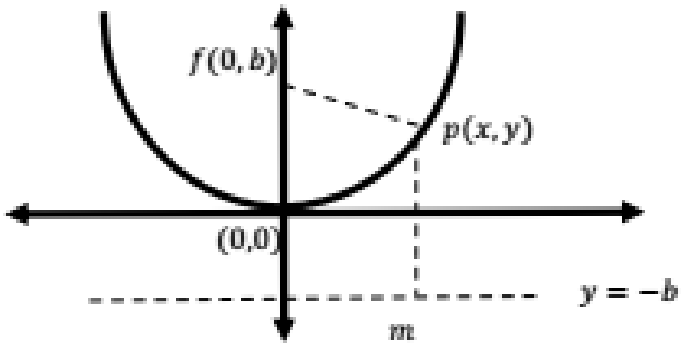
Different forms of parabola

- The vertex of the parabola is the origin (0,0) and its axis is the x-axis.

Shape of parabola		

Equation of parabola	$y^2 = -4a\,x$	$\boldsymbol{y^2 = 4a\,x}$
Direction	Negative side of x-axis	**Positive side of x-axis**
Vertex	$(0,0)$	$\boldsymbol{(0,0)}$
Focus	$(-a,0)$	$\boldsymbol{(a,0)}$
Directrix	$x = a$	$\boldsymbol{x = -a}$
Latus rectum	$4a$	$\boldsymbol{4a}$
Axes	$y = 0$	$\boldsymbol{y = 0}$

The equation of a parabola whose vertex is the origin and whose axis is the positive side of the y-axis



$\because\ pf = pm$

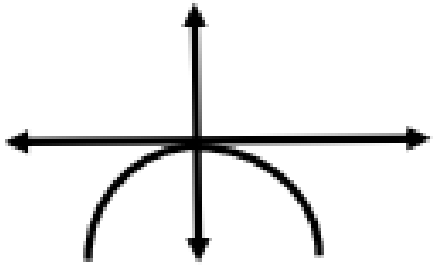
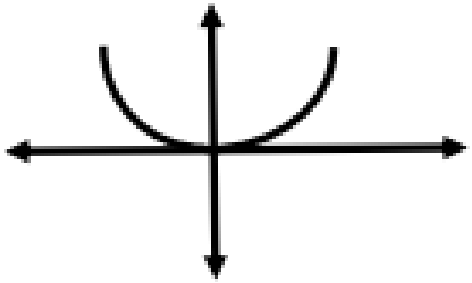
$$\sqrt{(x-0)^2+(y-b)^2} = y+b$$

$$(x-0)^2+(y-b)^2 = (y+b)^2$$

$$x^2 + y^2 - 2b\,y + b^2 = y^2 + 2b\,y + b^2$$

$$x^2 = 4b\,y$$

- The vertex of the parabola is the origin (0,0) and its axis is the y-axis.

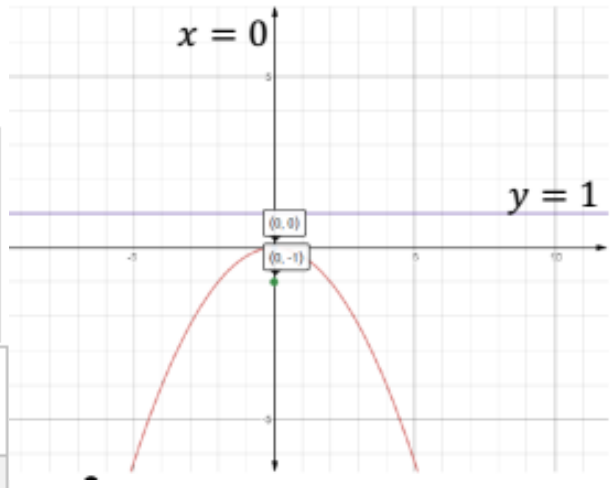
Shape of parabola		
Equation of parabola	$x^2 = -4b\,y$	$\boldsymbol{x^2 = 4b\,y}$
Direction	Negative side of y-axis	**Positive side of y-axis**
Vertex	$(0,0)$	$\boldsymbol{(0,0)}$
Focus	$(0,-b)$	$\boldsymbol{(0,b)}$
Directrix	$y = b$	$\boldsymbol{y = -b}$
Latus rectum	$4b$	$\boldsymbol{4b}$
Axes	$x = 0$	$\boldsymbol{x = 0}$

Example 1

Study the following parabola $2x^2 + 8y = 0$

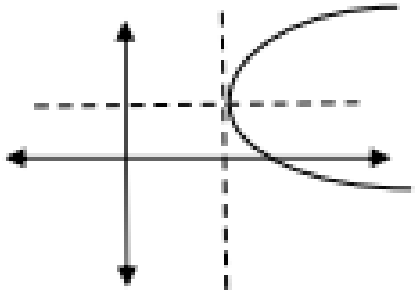
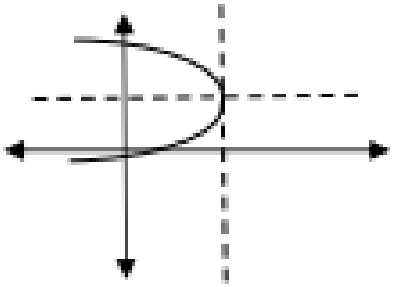
$\because\ 2x^2 + 8y = 0 \Rightarrow 2x^2 = -8y$

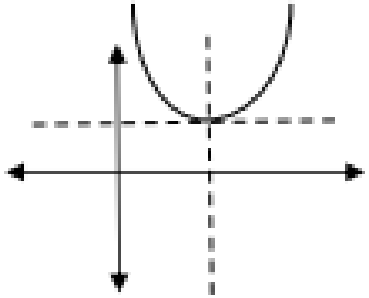
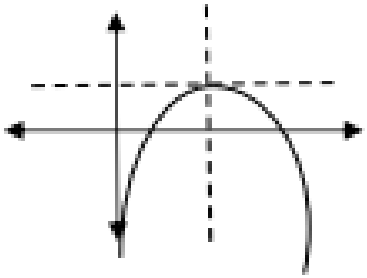
$\therefore\ x^2 = -4y \Rightarrow x^2 = -4by \Rightarrow b = 1$



Equation of parabola	$x^2 = -4by$	$x^2 = -4y$
Vertex	(0,0)	(0,0)
Focus	$(0, -b)$	$(0, -1)$
Directrix	$y = b$	$y = 1$
Latus rectum	$4b$	4
Axes	$x = 0$	$x = 0$

Forms of a parabola with a vertex (α, β)

Shape of parabola		
Equation of parabola	$(y-\beta)^2 = 4a\,(x-\alpha)$	$(y-\beta)^2 = -4a\,(x-\alpha)$
Direction	Positive side of x-axis	**Negative side of x-axis**
Vertex	(α, β)	(α, β)
Focus	$(\alpha + a, \beta)$	$(\alpha - a, \beta)$
Directrix	$x = \alpha - a$	$x = \alpha + a$
Latus rectum	$4a$	$4a$
Axes	$y = \beta$	$y = \beta$

Shape of parabola		
Equation of parabola	$(x-\alpha)^2 = 4b\,(y-\beta)$	$\boldsymbol{(x-\alpha)^2 = -4b\,(y-\beta)}$
Direction	Positive side of y-axis	**Negative side of y-axis**
Vertex	(α, β)	$\boldsymbol{(\alpha, \beta)}$
Focus	$(\alpha, \beta + b)$	$\boldsymbol{(\alpha, \beta - b)}$
Directrix	$y = \beta - b$	$\boldsymbol{y = \beta + b}$
Latus rectum	$4b$	$\boldsymbol{4b}$
Axes	$x = \alpha$	$\boldsymbol{x = \alpha}$

The general equation of a parabola whose axis is parallel to the x-axis (horizontal axis) is on the form

$$x = a\,y^2 + b\,y + c$$

$$\because x = a\,y^2 + b\,y + c$$

$$\therefore a\,y^2 + b\,y = x - c$$

$$\therefore a\left(y^2 + \frac{b}{a}\,y\right) = x - c$$

$$\therefore a\left(y + \frac{b}{2a}\right)^2 = x - c + \frac{b^2}{4a}$$

$$\therefore a\left(y + \frac{b}{2a}\right)^2 = x - c + \frac{b^2}{4a}$$

$$\therefore \left(y + \frac{b}{2a}\right)^2 = \frac{1}{a}\left(x - \left(c - \frac{b^2}{4a}\right)\right)$$

By comparing this equation with the standard form $(y-\beta)^2 = 4a\,(x-\alpha)$, we find that the vertex of the parabola is $\left(\left(c - \frac{b^2}{4a}\right), -\frac{b}{2a}\right)$

The general equation of a parabola whose axis is parallel to the y-axis (vertical axis) is on the form

$$y = a\,x^2 + b\,x + c$$

We will convert this form to the standard form

$$(x-\alpha)^2 = 4b\,(y-\beta)\,.$$

thank you